TRIBHUVAN UNIVERSITY

INSTITUTE OF SCIENCE AND TECHNOLOGY

HIMALAYA COLLEGE OF ENGINEERING

CHYASAL,LALITPUR

checked

Lab Report No:- 01

Title:- IMPLEMENTING APPLICATION OF MONTE CARLO METHOD.

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Submitted To:-

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LAB1: Implementing Application of Monte carlo method.

Objective:

To implement the Monte carlo method using c programming to estimate the probability of obtaining 3, 6 or 9 heads when a coin is tossed 10 times through repeated random simulations.

Theory:

The Monte carlo method is a computational technique that uses random sampling and repeated experiments to estimate numerical value. It is particularly useful for solving problems that involve probability, randomness & complex mathematical calculations

In this method we also calculate the error percentage. And also find the probability of coin toss.

$$\text{probability} = \frac{\text{Number of successful experiments}}{\text{Total Number of experiments}}$$

Algorithm for calculating value of PI(π) using monte carlo method.

1. initialize circle point, square point, interval to zero.
2. generate random point 'x' (between 0 & 1).
3. generate random point 'y' (between 0 & 1).
4. calculate, $d = r^2 + y^2$
5. If $d \leq 1$, increment circle point
6. Increment interval, increment square point.
7. if increment < no. of iteration repeat from step 2.
8. calculate, $pi = 4 * \frac{\text{circle-point}}{\text{square-point}}$
9. Display the value.

Theory of Activity 2?

Activity 1:

Q1.calculate the value of pi using monte carlo method

```
#include<stdio.h>
#include<math.h>
#include<conio.h>
#include<stdlib.h>
#define SEED 35791246
#define PI 3.14159265358979323846

int main()
{
  int N=0;
  double x,y;
  int i ,n=0;
  double z;
  double pi;
  double error_percentage;
  printf("Monte Carlo Method Lab1 Q1\n");
  printf("Enter the number of iterations used to estimate pi: ");
  scanf("%d",&N);
  srand(SEED);
  n=0;

  for(i=0;i<N;i++)
  {
    x = (double) rand() / RAND_MAX;
    y = (double) rand() / RAND_MAX;
    z=x*x+y*y;
```

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```
    if (z<=1)
        n++;
  }
  pi = (double) n / N * 4;
  error_percentage = fabs((pi - PI) / PI) * 100;
  printf("No of trials = %d, estimate of pi is %g\n", N, pi);
  printf("Error percentage: %.2f%%\n", error_percentage);
  printf("Name:Anil Pal Roll:5");
  getch();
}
```

```
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Monte Carlo Method Lab1 Q1
Enter the number of iterations used to estimate pi: 5
No of trials = 5, estimate of pi is 2.4
Error percentage: 23.61%
Name:Anil Pal Roll:5
Process returned 0 (0x0)   execution time : 3.951 s
Press any key to continue.
```

Q2: Calculate the Error Percentage and Repeat the experiment for at least 5 different value of input iteration and tabulate Actual pi,

```
#include <stdio.h>
#include <math.h>
#include <stdlib.h>
#include <time.h>
#define PI 3.1415926535897932384 6
double monte_carlo_pi(int iterations) {
  int inside_circle = 0;
```

```
    for (int i = 0; i < iterations; i++) {
        double x = (double)rand() / RAND_MAX;
        double y = (double)rand() / RAND_MAX;
        if (x*x + y*y <= 1.0) {
            inside_circle++;
        }
    }
    return 4.0 * inside_circle / iterations;
}

int main() {
    srand(time(NULL));

    int n;
    printf("Enter the number of different iteration values you want to try: ");
    scanf("%d", &n);
    int iterations[n];
    for (int i = 0; i < n; i++) {
        printf("Enter iteration value %d: ", i + 1);
        scanf("%d", &iterations[i]);
    }
    printf("\nS.N\tIterations\tEstimated Pi\tActual Pi\t%% Error\n");
    printf("-----------------------------------------------------------\n");

    for (int i = 0; i < n; i++) {
        double estimated_pi = monte_carlo_pi(iterations[i]);
        double error_percentage = fabs((estimated_pi - PI) / PI) * 100;

        printf("%d\t%d\t\t%.6f\t%.6f\t%.4f%%\n",
```

```
        i + 1, iterations[i], estimated_pi, PI, error_percentage);
    }

    return 0;
}
```

```
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Enter the number of different iteration values you want to try: 5
Enter iteration value 1: 100
Enter iteration value 2: 200
Enter iteration value 3: 1000
Enter iteration value 4: 2000
Enter iteration value 5: 5000

S.N     Iterations      Experiments Output      True Value      % Error
-----------------------------------------------------------------------
1       100             2.960000                3.141593        5.7803%
2       200             3.200000                3.141593        1.8592%
3       1000            3.168000                3.141593        0.8406%
4       2000            3.170000                3.141593        0.9042%
5       5000            3.158400                3.141593        0.5350%
Name:Anil Pal  Roll no:5
Process returned 0 (0x0)   execution time : 11.962 s
```

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Activity 2:

Q1:Use monte carlo method to calculate the probability of getting 3,6 or 9 heads in tossing of coin 10 times.

```
#include <stdio.h>

int main()
{
  int total_experiments, successful_experiments;
  float probability;

  printf("Enter total number of experiments: ");
  scanf("%d", &total_experiments);

  printf("Enter number of successful experiments (3,6 or 9 heads cases): ");
  scanf("%d", &successful_experiments);

  probability = (float)successful_experiments / total_experiments;

  printf("\nTotal Experiments = %d", total_experiments);
  printf("\nSuccessful Experiments = %d", successful_experiments);
  printf("\nEstimated Probability = %f", probability);

  printf("\n\nName: Anil Pal  Roll no:5");
  return 0;
}
```

```
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Enter total number of experiments: 1000
Enter number of successful experiments (3,6 or 9 heads cases): 250

Total Experiments = 1000
Successful Experiments = 250
Estimated Probability = 0.250000

Name: Anil Pal  Roll no:5
Process returned 0 (0x0)   execution time : 8.950 s
Press any key to continue.
```

DISCUSSION AND CONCLUSION

This is our first lab of simulation & modeling. In this experiment a C program was used to simulate the tossing of a coin multiple times. & in this lab we discussed about how we implement monte carlo method in C program to calculate pi value. The teacher help us to calculate the pi value & error percentage and probability of coin. This experiment demonstrates how the Monte carlo method can be used to approximate probabilities through repeated random trails.

Conclusion

In the last, we were able to calculate pi value, error percentage, probability using monte carlo method in C programming language.

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